

Apache Kafka & Confluent Kafka Administration

Course Outline

1. Duration

- 40 Hours

2. Objectives

At the end of this workshop, participants will be able to:

- Understand messaging concepts and core Apache Kafka architecture, components, and distributed design principles.
- Install, configure, and administer Apache Kafka clusters, including storage, retention, cluster operations, and high availability.
- Produce and consume messages using both CLI tools and Java applications on Apache Kafka.
- Transition core administration skills to the Confluent Kafka platform and understand how it extends Apache Kafka.
- Administer advanced Confluent Kafka capabilities — security (RBAC), monitoring, and performance tuning — using Confluent's key administration features.
- Troubleshoot Confluent Kafka clusters, diagnose and prevent message loss, and evaluate migration considerations from Apache Kafka to Confluent Kafka.

3. Audience

- Kafka Administrators, Linux Administrators, OpenShift Administrators
- Application Developers (Java)
- Platform Engineers / DevOps Engineers supporting Kafka-based messaging

4. Prerequisites

- Understanding of Messaging Concepts (IBM MQ or similar messaging systems)
- Exposure to Distributed Systems / OpenShift Container Platform
- Basic Linux Administration

5. Hardware & Software Requirements

- Desktop/Laptop with minimum 16 GB RAM
- Open Internet connection (minimum 10 Mbps per user)
- Local Admin Access
- Git client, SSH Client
- Docker Desktop
- VS Code with GitHub Copilot
- Java 21 / Python 12
- Shared Apache Kafka cluster access (AWS Cloud)
- Shared Confluent Kafka cluster access

6. Outline

Module 1 — Messaging & Kafka Fundamentals

Topics

- Messaging concepts recap: queues vs. pub-sub, comparison with IBM MQ and traditional messaging systems
- Kafka overview and use cases
- Kafka ecosystem: Producers, Consumers, Brokers, Topics, Partitions
- Kafka architecture: cluster design, offsets, replicas
- Message retention and compaction
- ZooKeeper vs. KRaft mode — architecture overview

Hands-on

- Docker Desktop and local environment setup
- Install and run Apache Kafka locally
- Explore Kafka CLI tools

✓ **Objective Alignment:** *Understand messaging concepts and core Apache Kafka architecture, components, and distributed design principles.*

Module 2 — Kafka Installation, Setup & CLI Operations

Topics

- System requirements and installation options (Linux, Docker, KRaft mode)
- Kafka CLI tools overview
- Creating, listing, describing, and deleting topics
- Producing and consuming messages using CLI

Hands-on

- Multi-broker local cluster setup
- Topic create / list / describe / delete via CLI
- Produce and consume test messages

✓ **Objective Alignment:** *Install, configure, and perform core administration on Apache Kafka clusters.*

Module 3 — Cluster Configuration, Storage & Retention

Topics

- Broker configuration parameters
- Log directories and segment management
- Topic-level configuration overrides
- Retention policies (time-based and size-based); cleanup policies: delete vs. compact
- min.insync.replicas and acknowledgments (acks)
- Storage optimization techniques

Hands-on

- Configure retention and replication on the shared Apache Kafka cluster (AWS)
- Test message durability under different acks settings

✓ **Objective Alignment:** *Administer storage, retention, and configuration on Apache Kafka clusters.*

Module 4 — Producing & Consuming Messages

Topics

- Producer configuration tuning (batch.size, linger.ms, retries, acks, idempotence)
- Consumer groups, offset management, commit strategies, and rebalancing
- Delivery semantics: at-most-once, at-least-once, exactly-once
- Writing Kafka producer and consumer applications in Java
- Error handling and retry patterns

Hands-on

- Build Java producer and consumer applications in VS Code (with GitHub Copilot)
- Run applications against the shared Apache Kafka cluster on AWS

✓ **Objective Alignment:** *Administer and validate message flow by producing and consuming via CLI and Java applications on Apache Kafka.*

Module 5 — Cluster Operations, Replication & High Availability

Topics

- Adding and removing brokers; cluster balancing and partition reassignment (kafka-reassign-partitions.sh)
- ISR (In-Sync Replicas), leader election, and handling broker failures
- Rolling restarts and upgrade considerations
- Cluster capacity planning

Hands-on

- Partition reassignment exercise on the shared Apache Kafka cluster
- Simulate broker failure and recovery

✓ **Objective Alignment:** *Complete core Apache Kafka administration: cluster operations and high availability.*

Module 6 — Introducing Confluent Kafka: Platform, Architecture & Administration Basics

Topics

- Confluent Platform overview and architecture
- Confluent vs. Apache Kafka: feature and licensing comparison
- Confluent Cloud vs. self-managed Confluent Platform — deployment models
- Confluent CLI overview and authentication
- Confluent Control Center walkthrough
- Topic management via Confluent CLI and Control Center

Hands-on

- Access and explore the shared Confluent Kafka cluster / Confluent Cloud console
- Topic create / list / describe / delete on the shared Confluent Kafka cluster via CLI and Control Center

✓ **Objective Alignment:** *Transition core Kafka administration skills to the Confluent Kafka platform.*

Module 7 — Administering Kafka Security

Topics

- Confluent RBAC (Role-Based Access Control)
- API keys and secrets management (Confluent Cloud)
- Confluent Platform authentication and authorization overview
- Encryption and network security in Confluent-managed environments
- Comparing the Confluent security model to Apache Kafka ACLs, SASL, and SSL

Hands-on

- Configure RBAC roles and API keys on the shared Confluent Kafka cluster
- Validate access with test users / service accounts

✓ **Objective Alignment:** *Secure Confluent Kafka clusters using Confluent's administration and access-control model.*

Module 8 — Monitoring & Performance Tuning with Confluent Kafka

Topics

- Confluent Control Center monitoring and alerting
- Confluent Metrics API and Health+ overview
- Cluster load metrics: throughput and latency indicators
- Performance considerations and scaling in Confluent-managed clusters
- Comparing to Apache Kafka JMX / Prometheus-based monitoring

Hands-on

- Monitor the shared Confluent Kafka cluster via Control Center and the Metrics API
- Set up an alert and analyze consumer lag

✓ **Objective Alignment:** *Monitor and tune Confluent Kafka clusters for performance using Confluent's administration tooling.*

Module 9 — Troubleshooting & Preventing Message Loss on Confluent Kafka

Topics

- Common Confluent Cloud / Platform errors and log analysis
- Root causes of message loss: acks configuration, replication factor, quota and throttling limits
- Troubleshooting connectivity and client-side issues
- Consumer lag analysis and remediation in Confluent
- Confluent support and escalation model
- Best-practice checklist for preventing message loss

Hands-on

- Diagnose and resolve simulated message-loss and connectivity scenarios on the shared Confluent Kafka cluster

✓ **Objective Alignment:** *Troubleshoot Confluent Kafka clusters and diagnose, resolve, and prevent message loss and other production issues.*

Module 10 — Ecosystem, Migration, Backup/DR & Capstone

Topics

- Confluent managed connectors overview (source/sink)
- Confluent Schema Registry: schema evolution and compatibility
- ksqlDB on Confluent — overview
- Backup and DR: Confluent Replicator, multi-datacenter replication
- Migration considerations: Apache Kafka → Confluent Kafka (data migration, client cutover, dual-write strategies)

Hands-on / Capstone

- Explore managed connectors and Schema Registry on the shared Confluent Kafka cluster
- End-to-end capstone: administer, monitor, and troubleshoot a scenario on the Confluent Kafka cluster
- Guided review and Q&A

✓ **Objective Alignment:** *Administer the Confluent Kafka ecosystem and evaluate migration considerations from Apache Kafka to Confluent Kafka.*